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Caracterizaciones de la convergencia casi segura

Observación 1. Veamos varias caracterizaciones útiles de la convergencia casi segura:

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\begin{aligned}
X_n \xrightarrow[n\to\infty]{\text{c.s.}} X \ &\overset{\substack{\text{defn}\\ \downarrow}}{\iff} \exists A \in \Sigma : \mathbb{P}\left(A\right) = 0 \wedge \forall \omega \in A^{\mathsf{c}} : \lim_{n\to\infty} X_n(\omega) = X(\omega) \\
&\overset{\substack{\text{defn de límite en } \mathbb{R}\\ \downarrow}}{\iff} \exists A \in \Sigma : \mathbb{P}\left(A\right) = 0 \\
&\qquad\qquad \wedge \forall \omega \in A^{\mathsf{c}} : \forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall n \geq N : |X_n(\omega) - X(\omega)| < \varepsilon \\
&\overset{\substack{\texttt{liminf-con/1}\\ \downarrow}}{\iff} \exists A \in \Sigma : \mathbb{P}\left(A\right) = 0 \wedge \forall \varepsilon > 0 : A^{\mathsf{c}} \subset \liminf_{n\to\infty} \left\{|X_n - X| < \varepsilon\right\} \\
&\iff \forall \varepsilon > 0 : \mathbb{P}\left(\liminf_{n\to\infty} \left\{|X_n - X| < \varepsilon\right\}\right) = 1 \\
&\overset{\substack{\texttt{ejer-limsup-liminf-con/1:ejer-limsup-liminf-con/(b)}\\ \downarrow}}{\iff} \forall \varepsilon > 0 : \mathbb{P}\left(\limsup_{n\to\infty} \left\{|X_n - X| > \varepsilon\right\}\right) = 0 \\
&\overset{\substack{\text{Fatou}\\ \downarrow}}{\iff} \forall \varepsilon > 0 : \limsup_{n\to\infty} \mathbb{P}\left(|X_n - X| > \varepsilon\right) = 0.
\end{aligned}
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